

WATER QUALITY MONITORING SYSTEM USING IOT SENSORS AND MACHINE LEARNING

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ABSTRACT

This study develops an IoT-based water quality monitoring system with SVM classification for detecting potability, achieving 94.2% classification accuracy and enabling real-time alerts for unsafe water conditions.

INTRODUCTION

Access to clean water is a persistent challenge in coastal Cavite communities. Manual water quality testing is infrequent and slow. IoT-based continuous monitoring with automated classification can provide real-time safety alerts. This study deploys a multi-sensor IoT system in Bacoar Bay's water supply outlet and trains an SVM classifier to predict water potability.

METHODOLOGY

Sensors deployed: pH sensor, turbidity sensor (TSD-10), dissolved oxygen (DO) sensor, and temperature probe, all connected to ESP32 microcontroller with ThingSpeak cloud logging. 6 months of 15-minute interval sensor data (17,280 readings) were collected. SVM with RBF kernel was trained on labeled samples (WHO potability standards as ground truth). Features: pH, turbidity (NTU), DO (mg/L), temperature (°C). Evaluation: 5-fold cross-validation accuracy, precision, recall.

RESULTS

SVM achieved 94.2% classification accuracy, precision 93.8%, recall 94.6%. Turbidity and pH were the most discriminative features (ANOVA F-test). The system correctly identified 3 contamination events during the monitoring period that were later confirmed by LGU testing. Alert response time (sensor reading to SMS notification) averaged 12 seconds.

DISCUSSION

The IoT-SVM combination provides actionable, real-time water safety monitoring feasible for deployment in coastal communities. SVM's lightweight inference (0.001s) is ideal for edge deployment on ESP32. pH sensor drift was observed at 3-month intervals, necessitating regular recalibration. Future work should evaluate neural network models and extend sensor array with E. coli detection modules.